Hotel Booking Cancellation Prediction

Project report for review · Data science project

Hotel bookings: the data is cleaned, explored and split; model training is the next stage.

1. Aim

Predict from booking details whether a reservation will be cancelled (**is_canceled**: 1 = cancelled, 0 = kept). A quick tool like this helps a hotel plan for overbooking, pricing and staffing.

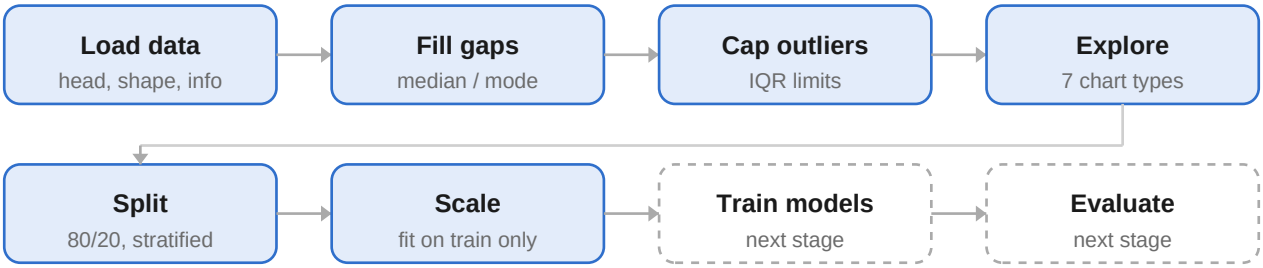
2. Data

Each row of `hotel_bookings.csv` is one booking. There are **119,390** bookings and **32** columns from a City Hotel and a Resort Hotel (arrivals 2015–2017). 20 columns are numbers and 12 are text, so the text columns still need encoding before modelling.

Column group	Examples	Meaning
Booking	hotel, lead_time, arrival date, stays, deposit_type	When, where and how the room was booked
Guests	adults, children, babies, country, customer_type	Who is staying
History	is_repeated_guest, previous_cancellations	What the guest did before
Sales	market_segment, distribution_channel, agent, adr	How it was sold and the daily price
Extras	total_of_special_requests, parking, booking_changes	Requests and changes made
Target	is_canceled	The answer to predict (37.04% = 1)

3. Workflow at a glance

Data preparation is finished; modelling is next



Shaded = completed · dashed = planned

project workflow · 8 steps

Shaded boxes are finished; dashed boxes are planned.

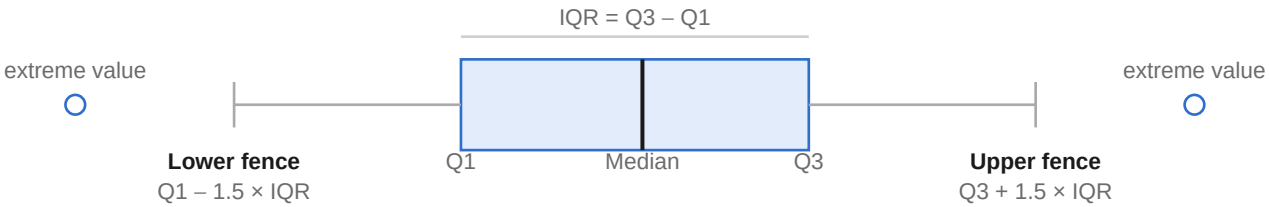
4. Data cleaning

- **Missing values:** children (4), country (488), agent (16,340) and company (112,593) had gaps. Numbers were filled with the column median and text with the most common value. No rows were dropped.
- **Duplicates:** checked with `duplicated().sum()`, which found 31,994 repeated rows. They were reported but not yet removed.
- **Summary statistics:** `describe()` gave the count, mean, spread and range of every number column.

5. Outliers

An outlier is a value far from the rest. I flagged values beyond 1.5 × IQR from the middle 50% of the data, capped them at that fence instead of deleting rows, and re-ran the check. For example, the longest lead time dropped from 737 to 373 days.

Capping pulls extreme values back to the fence instead of deleting them



IQR capping · illustration, not project data

Capping is a design choice, and here it went too far: for columns where most values are 0 (adults, children, babies, booking_changes and similar), the fence sits at 0, so every value was flattened to a constant. See limitations.

6. Exploratory analysis

Chart	What it shows
Histograms of all numeric features	How values are spread
Box plot of numeric features	Median, spread and leftover outliers
Count plot of is_canceled	Whether kept and cancelled classes are balanced
Scatter: lead_time vs adr	Whether lead time and price move together
Box plots of lead_time and adr	Spread of the two main features
Bar chart by hotel type	Cancellations at City vs Resort Hotel
Correlation heatmap	How strongly every pair of features is related

What the charts showed:

- **Balance:** 62.96% kept and 37.04% cancelled, so the classes are moderately imbalanced.
- **Hotel:** the City Hotel has more bookings and a higher cancellation share than the Resort Hotel.
- **Lead time:** longer lead times go with more cancellations (correlation about +0.3).
- **Special requests:** more requests go with fewer cancellations (about –0.3).

7. Preparing for modelling

- **Inputs and target:** the 31 other columns form X; `is_canceled` is y.
- **Split:** 80% training (95,512 rows), 20% testing (23,878 rows). `stratify=y` keeps the cancelled share equal in both parts, and `random_state=42` makes the split reproducible.

- **Scaling:** `StandardScaler` puts number columns on one scale (mean 0, standard deviation 1), since lead time runs into hundreds while adults is 1 or 2. It is fitted on training data only, so no test information leaks in.

8. Next steps

- **Fix first:** drop `reservation_status` and its date, remove duplicates, and cap only continuous columns (lead_time, adr).
- **Encode:** turn the 12 text columns into numbers with one-hot encoding inside a scikit-learn pipeline.
- **Models:** Logistic Regression baseline, Decision Tree, Random Forest, Gradient Boosting.
- **Metrics:** accuracy, precision, recall, F1, confusion matrix and ROC-AUC, with extra attention to cancellations the model misses.
- **Refinement:** cross-validation, hyperparameter tuning and feature importance.

9. Limitations

- **Target leakage:** `reservation_status` (Canceled / Check-Out / No-Show) is still in X and gives the answer away. It must be removed before any model is trained.
- **Flattened columns:** after capping, 9 columns (adults, children, babies, is_repeated_guest, previous_cancellations, previous_bookings_not_canceled, booking_changes, days_in_waiting_list, parking) have no variation, and company is fixed at 179. This is why the heatmap has blank rows.
- **Company column:** 94% is missing, so filling it with the median is not meaningful.
- **Order of steps:** medians and IQR limits were computed before the split, a mild form of data leakage; a pipeline would fix this.
- **Leftover problems:** some negative adr values remain, and duplicates are still in the data.
- Results depend on how the bookings were collected, and the data is only from 2015–2017.

10. Conclusion

The raw dataset is now loaded, explored and split with a leak-free scaling step. Before training, the leakage and capping problems above need fixing and the text columns need encoding. Then comes training and comparing models to predict which bookings will be cancelled.